Custom Cat command: It’s code and working by 24JE0611

```
d_viva@d-viva-18:~/OS_Projects_G5/Project_2/custom_commands$ ls
custom_cat.c  custom_grep.c  sample_cat.txt  sample.txt
d_viva@d-viva-18:~/OS_Projects_G5/Project_2/custom_commands$ gcc custom_cat.c -o custom_cat
d_viva@d-viva-18:~/OS_Projects_G5/Project_2/custom_commands$ cd ..
d_viva@d-viva-18:~/OS_Projects_G5/Project_2$ ls
custom_commands  Documentations  README.md  Shell
d_viva@d-viva-18:~/OS_Projects_G5/Project_2$ cd Shell/
d_viva@d-viva-18:~/OS_Projects_G5/Project_2/Shell$ ls
shell.c
d_viva@d-viva-18:~/OS_Projects_G5/Project_2/Shell$ gcc shell.c -o shell
d_viva@d-viva-18:~/OS_Projects_G5/Project_2/Shell$ ./shell
G4_Shell> G4_Shell> ../custom_commands/custom_cat ../custom_commands/sample_cat.txt
Command Not Found!!
G4_Shell> ./../custom_commands/custom_cat ../custom_commands/sample_cat.txt
Hello Davey!
This is a test file for the OS Project 2.
Implementation: custom_cat
Status: Minimalist and Functional.
IIT (ISM) Dhanbad - CSE
G4_Shell> exit
G4_Shell>
G4_Shell> ^C
```

```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#define BUFFER_SIZE 4096

int main(int argc, char *argv[]) {
    // Check if the user provided a filename
    if (argc < 2) {
        printf("Usage: %s <filename>\n", argv[0]);
        return 1;
    }
    // Open the file in read-only mode
    int fd = open(argv[1], O_RDONLY);
    if (fd == -1) {
        perror("Error opening file");
        return 1;
    }
    char buffer[BUFFER_SIZE];
    ssize_t bytesRead;
    // Read from file and write to standard output (stdout = 1)
    while ((bytesRead = read(fd, buffer, BUFFER_SIZE)) > 0) {
        if (write(1, buffer, bytesRead) != bytesRead) {
            perror("Error writing to stdout");
            close(fd);
            return 1;
        }
    }
    if (bytesRead == -1) {
        perror("Error reading file");
    }

    close(fd);
    return 0;
}
```